Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 60%. We keep your highest score.

Next item →

1. **What is a key requirement of the three-layer MCP architecture when an application needs to connect to multiple servers?** 1 / 1 point
   - A single host-client pair automatically scales across all servers
   - The host process must create one MCP client instance per server connection
   - Multiple hosts must be deployed to manage each server
   - A single MCP client must multiplex all server connections

   Nice work

   MCP architecture requires a separate client instance for every server the host connects to.

2. **Which statement best describes the three mandatory phases of an MCP connection?** 1 / 1 point
   - Registration, execution, and teardown
   - Initialization, operation, and shutdown
   - Handshake, messaging, and audit logging
   - Initialization, discovery, and persistence

   Nice work

   MCP connections always follow initialization with capability negotiation, then operation, and finally shutdown with cleanup.

3. **What is the role of JSON-RPC 2.0 in MCP communications?** 1 / 1 point
   - It is used only for tool invocation messages between server and client
   - It defines requests, responses, and notifications to enable bidirectional client-server messaging
   - It ensures that every message sent must include a response
   - It is only required when MCP runs over HTTP transport

   Nice work

   JSON-RPC 2.0 provides the structure for all MCP messages, supporting full two-way communication.

4. **What does the @mcp.tool() decorator do when defining tools such as echo or write_file?** 1 / 1 point
   - It registers the tool but requires manual JSON schema configuration
   - It launches the MCP server over STDIO
   - It exposes the Python function as a callable tool and auto-generates its JSON schema from type hints
   - It limits the tool to reading only resource files

   Nice work

   The decorator makes the function callable by clients and builds the schema automatically.

5. **How do MCP resource templates such as file://resources/{filename} function?** 1 / 1 point
   - They return tool execution results such as echo output
   - They define URI patterns that allow clients to request resource content by substituting parameter values
   - They store rendered prompt templates for later LLM use
   - They automatically generate Python functions for file handling

   Nice work

   Clients supply a concrete URI, and the server returns the matching resource content.